IPR 1 Decision Brief

Purpose: Confirm Adelman interpretation, ISE 631 course connection, and roadmap for completing project requirements.

Requested decisions

- Adelman interpretation?
- Right LP themes: simplex, duality, complementary slackness, reduced costs, column generation?
- Roadmap satisfies the project prompt?
- Surveillance-pricing extension?

Agenda today

- Project scope and narrative
- Central research questions
- Adelman technique and LP connections
- Computational verification plan
- Roadmap to final submission

Successful IPR 1 end state

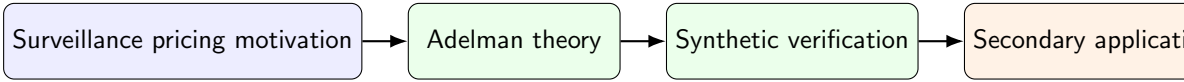
- Scope and questions approved
- LP theory emphasis approved
- Verification requirements approved
- Surveillance-pricing guardrails approved
- Milestones to final submission approved

Project Scope: Core Assignment with Contemporary Motivation

Working Project Concept

Core: Understand, explain, and computationally verify Adelman's dynamic bid-price methodology through the lens of linear programming.

Extension: Investigate whether Adelman's framework can be meaningfully applied to a surveillance-pricing setting using real or realistically calibrated data.



Scope Guardrail

Core project success does **not** depend on obtaining ideal surveillance-pricing data. If the application is infeasible, the completed project remains: Adelman explanation, LP-theory connection, computational verification, and critique.

Approved Central Research Questions

1. What is gained—and lost—by replacing the exact MDP value function with Adelman's approximate value function?
2. Why does Adelman's approximate LP produce a bound that dominates the deterministic/fluid formulation?
3. How do LP duality, reduced costs, and column generation make the resulting large-scale problem computationally tractable?
4. How does the method scale with increasing state/action-space complexity, and can portions of the problem be truncated without materially changing the solution?
5. How large are the performance gaps among the fluid formulation, approximate LP, dynamic bid-price policy, and exact MDP—and under what conditions do dynamic prices provide the greatest benefit?
6. Can Adelman's framework be meaningfully applied to a surveillance-pricing problem using real or realistically calibrated data?

IPR 1 Decision

Adelman's Technique: What I Need to Explain Correctly

Exact MDP LP

- State includes remaining time and capacity vector.
- Bellman inequalities create an LP over value functions.
- Exact formulation is theoretically clean but computationally intractable at scale.

Approximate LP

- Replace exact value function with structured affine approximation.
- Preserve marginal value logic while reducing representational burden.

Policy Logic

- Resource capacity has opportunity cost.
- Bid prices approximate marginal value of remaining capacity.
- Accept request when revenue exceeds implied resource opportunity cost.

Professor Feedback Needed

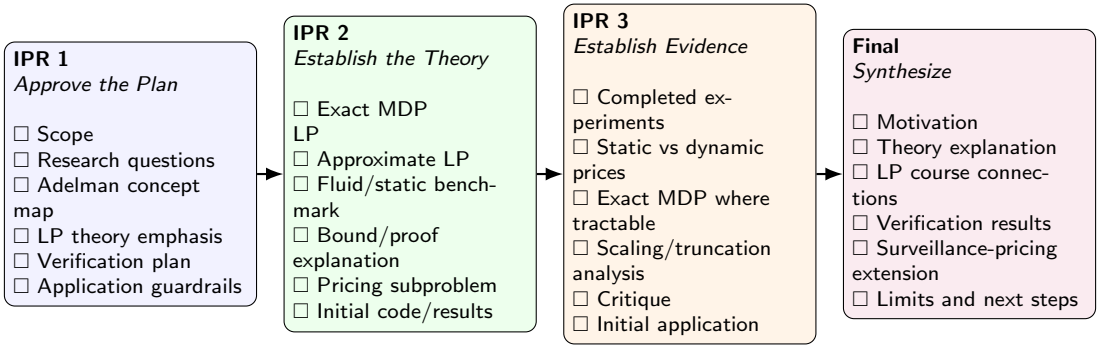
- Which proof/result must I understand deeply?
- What level of derivation is expected in final brief?
- What is sufficient for "verification" versus full

Connection to ISE 631: LP Themes to Emphasize

Course Theme	Planned Connection to Adelman
Primal/dual LP	Exact MDP LP has value-function variables; dual variables can be interpreted as state-action occupancy or resource valuation.
Complementary slackness	Links optimal policies to active constraints/support of the dual solution.
Shadow prices	Bid prices are operational interpretations of marginal resource values.
Simplex/reduced costs	Pricing logic asks whether omitted variables/constraints can improve or violate current solution.
Column generation	Solves a restricted problem, uses dual information to find violated constraints or improving columns, and iterates.
Approximation	Replaces exact, intractable LP with structured value-function approximation while preserving decision-relevant marginal values.

IPR 1 Discussion Question

Glidepath: Deliverables by IPR



Computational Verification and Application Plan

Verification Plan

Compare performance across synthetic instances:

- deterministic/fluid formulation;
- static bid-price policy;
- Adelman dynamic bid-price policy;
- exact MDP solution where tractable.

Vary capacity tightness, demand uncertainty, resource/product complexity, and horizon length.

Measures of Interest

- objective-value gaps;
- policy performance;

Surveillance-Pricing Extension

Feasibility test: can the application be framed as sequential decision-making under scarce resources and uncertain future demand?

- state;
- action;
- reward/revenue;
- capacity/resource constraint;
- demand model;
- transition logic.

Close IPR 1 With These Decisions